

Networking

1- create vpc

```
[janna@myhost terrlab1]$ terraform apply
```

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_vpc.myvpc will be created
+ resource "aws_vpc" "myvpc" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                               = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id               = (known after apply)
  + owner_id                         = (known after apply)
  + region                            = "us-east-1"
  + tags                              = {
    + "Name" = "Network_vpc"
  }
  + tags_all                          = {
    + "Name" = "Network_vpc"
  }
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.myvpc: Creating...

aws_vpc.myvpc: Creation complete after 4s [id=vpc-0a7f5e6a893982551]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Your VPCs (2) Info								
Find VPCs by attribute or tag								
Last updated 1 minute ago Actions Create VPC								
☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	Network_vpc	vpc-0a7f5e6a893982551	Available	-	-	Off	10.0.0.0/16	-
☐	-	vpc-05721e3ac3b24c669	Available	-	-	Off	172.31.0.0/16	-

2- create internet gateway

```
[janna@myhost terrlab1]$ terraform apply
aws_vpc.myvpc: Refreshing state... [id=vpc-0a7f5e6a893982551]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_internet_gateway.gw will be created
+ resource "aws_internet_gateway" "gw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "us-east-1"
  + tags     = {
    + "Name" = "IN_terr"
  }
  + tags_all = {
    + "Name" = "IN_terr"
  }
  + vpc_id   = "vpc-0a7f5e6a893982551"
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_internet_gateway.gw: Creating...

aws_internet_gateway.gw: Creation complete after 2s [id=igw-055411295b4aab78c]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Internet gateways (2) [Info](#)

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	IN_terr	igw-055411295b4aab78c	Attached	vpc-0a7f5e6a893982551 Network_vpc	531472034848
☐	-	igw-0ec1b5616bcf4525d	Attached	vpc-05721e3ac3b24c669	531472034848

-Create 2 subnets private and 2 public

NAME	PROVIDER ID	STATE	REGION	AVAILABILITY ZONE	IPV4 CIDR BLOCK	IPV6 CIDR BLOCK	ROUTE TABLE ID
☐ -	subnet-08a89210c9a0b8aba	Available	vpc-05721e3ac3b24c669	Off	172.31.48.0/20	-	-
☐ public2	subnet-00bd11a13c1e5cf04	Available	vpc-0a7f5e6a893982551 Net...	Off	10.0.2.0/24	-	-
☐ -	subnet-0c8c648bd9f58dcdf	Available	vpc-05721e3ac3b24c669	Off	172.31.64.0/20	-	-
☐ -	subnet-0fab2a4f9be675c8	Available	vpc-05721e3ac3b24c669	Off	172.31.0.0/20	-	-
☐ -	subnet-0e632a5789dca6a4d	Available	vpc-05721e3ac3b24c669	Off	172.31.32.0/20	-	-
☐ public1	subnet-0a43a8db4778da223	Available	vpc-0a7f5e6a893982551 Net...	Off	10.0.1.0/24	-	-
☐ -	subnet-02e49129d9c11c25f	Available	vpc-05721e3ac3b24c669	Off	172.31.80.0/20	-	-
☐ -	subnet-0642533982f157f87	Available	vpc-05721e3ac3b24c669	Off	172.31.16.0/20	-	-
☐ private1	subnet-00d6ece0ee236124e	Available	vpc-0a7f5e6a893982551 Net...	Off	10.0.4.0/24	-	-
☐ private1	subnet-0e22020712bcb3b68	Available	vpc-0a7f5e6a893982551 Net...	Off	10.0.3.0/24	-	-

3- create public route table/create public route/attach to public subnet

Route tables (3) Info

Last updated 2 minutes ago

Actions

Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0e187241f2e67ec62	-	-	Yes	vpc-0a7f5e6a893982551 Net...	531472034848
☐	-	rtb-08b4ac26a331fd222	-	-	Yes	vpc-05721e3ac3b24c669	531472034848
☐	public_rt	rtb-0291fd6daf5c40884	-	-	No	vpc-0a7f5e6a893982551 Net...	531472034848

Terraform will perform the following actions:

```
# aws_route.public_internet_route will be created
+ resource "aws_route" "public_internet_route" {
  + destination_cidr_block = "0.0.0.0/0"
  + gateway_id             = "igw-055411295b4aab78c"
  + id                    = (known after apply)
  + instance_id           = (known after apply)
  + instance_owner_id     = (known after apply)
  + network_interface_id  = (known after apply)
  + origin                 = (known after apply)
  + region                = "us-east-1"
  + route_table_id        = "rtb-0291fd6daf5c40884"
  + state                  = (known after apply)
}

# aws_route_table_association.public1_association will be created
+ resource "aws_route_table_association" "public1_association" {
  + id              = (known after apply)
  + region          = "us-east-1"
  + route_table_id = "rtb-0291fd6daf5c40884"
  + subnet_id      = "subnet-0a43a8db4778da223"
}

# aws_route_table_association.public2_association will be created
+ resource "aws_route_table_association" "public2_association" {
  + id              = (known after apply)
  + region          = "us-east-1"
  + route_table_id = "rtb-0291fd6daf5c40884"
  + subnet_id      = "subnet-00bd11a13c1e5cf04"
}
```

rtb-0291fd6daf5c40884 / public_rt

Actions

Details Info

Route table ID

rtb-0291fd6daf5c40884

VPC

vpc-0a7f5e6a893982551 | Network_vpc

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-055411295b4aab78c	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

4- create private route table/attach to private subnet

```
# aws_route_table.private_rt will be created
+ resource "aws_route_table" "private_rt" {
  + arn                = (known after apply)
  + id                 = (known after apply)
  + owner_id           = (known after apply)
  + propagating_vgws   = (known after apply)
  + region             = "us-east-1"
  + route              = (known after apply)
  + tags               = {
    + "Name" = "private_rt"
  }
  + tags_all           = {
    + "Name" = "private_rt"
  }
  + vpc_id              = "vpc-0a7f5e6a893982551"
}

# aws_route_table_association.private1_association will be created
+ resource "aws_route_table_association" "private1_association" {
  + id                 = (known after apply)
  + region             = "us-east-1"
  + route_table_id     = (known after apply)
  + subnet_id          = "subnet-0e22020712bcb3b68"
}

# aws_route_table_association.private2_association will be created
+ resource "aws_route_table_association" "private2_association" {
  + id                 = (known after apply)
  + region             = "us-east-1"
  + route_table_id     = (known after apply)
  + subnet_id          = "subnet-00d6ece0ee236124e"
}
```

rtb-06a31c6ac670501e5 / private_rt

Actions

Details [info](#)

Route table ID
[rtb-06a31c6ac670501e5](#)

VPC
[vpc-0a7f5e6a893982551](#) | [Network_vpc](#)

Main
[No](#)

Owner ID
[531472034848](#)

Explicit subnet associations
[2 subnets](#)

Edge associations
-

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (1)

[Filter routes](#)

Both [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Compute

7- create security group wh ich allow ssh from 0.0.0.0/0

```
resource "aws_security_group" "allow-ssh" {
  name        = "allow_ssh"
  description = "Allow TLS inbound traffic and all outbound traffic"
  vpc_id      = aws_vpc.myvpc.id

  ingress {
    description = "SSH"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    description = "Allow all outbound"
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "SG1"
  }
}
```

Security Groups (4) Info							
Find security groups by attribute or tag							
☐	Name	Security group ID	Security group name	VPC ID	Description	Owner	
☐	-	sg-04296d12fda9f90fb	default	vpc-0a7f5e6a893982551	default VPC security group	531472034848	
☐	-	sg-0c9419f04a694c1eb	default	vpc-05721e3ac3b24c669	default VPC security group	531472034848	
☐	SG1	sg-07993cb653771505d	allow_ssh	vpc-0a7f5e6a893982551	Allow TLS inbound traffic and all outbo...	531472034848	
☐	-	sg-0c4831576e59453ae	launch-wizard-1	vpc-05721e3ac3b24c669	launch-wizard-1 created 2026-04-03T1...	531472034848	

8- create security group that allow ssh and port 3000 from vpc cidr only

```
resource "aws_security_group" "allow-ssh-port_3000" {
  name      = "allow_ssh and port 3000 only"
  description = "Allow TLS inbound traffic and all outbound traffic"
  vpc_id    = aws_vpc.myvpc.id

  ingress {
    description = "SSH"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [aws_vpc.myvpc.cidr_block]
  }

  ingress {
    to_port     = 3000
    from_port   = 3000
    protocol    = "tcp"
    cidr_blocks = [aws_vpc.myvpc.cidr_block]
  }

  egress {
    description = "Allow all outbound"
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "SG2"
  }
}
```

sg-0c6cbe91e5f1bce02 - allow_ssh and port 3000 only

Actions

Details

Security group name

allow_ssh and port 3000 only

Security group ID

sg-0c6cbe91e5f1bce02

Description

Allow TLS inbound traffic and all outbound traffic

VPC ID

[vpc-0a7f5e6a893982551](#)

Owner

531472034848

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-082c60db96050f3a5	IPv4	Custom TCP	TCP	3000	10.0.0.0/16	-
☐	-	sgr-0f5b9554bba09b6fb	IPv4	SSH	TCP	22	10.0.0.0/16	SSH

7- create ec2(bastion) in public subnet with security group from 7

```
bastion.tf
1
2 resource "aws_instance" "bastion" {
3     ami                    = "ami-0c02fb55956c7d316"
4     instance_type         = "t3.micro"
5     subnet_id             = aws_subnet.public1.id
6     vpc_security_group_ids = [aws_security_group.allow-ssh.id]
7     associate_public_ip_address = true
8     key_name              = "terr"
9
10    tags = {
11        Name = "bastion"
12    }
13 }
```

Instance summary for i-0832733fd8f8b465b (bastion) [Info](#)

Updated less than a minute ago

Instance ID
i-0832733fd8f8b465b

IPv6 address
-

Hostname type
IP name: ip-10-0-1-109.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
44.200.53.215 [Public IP]

IAM role
-

IMDSv2
Optional
 EC2 recommends setting IMDSv2 to required | [Learn more](#)

Operator
-

Public IPv4 address
 44.200.53.215 | [open address](#)

Instance state
 Running

Private IP DNS name (IPv4 only)
 ip-10-0-1-109.ec2.internal

Instance type
t3.micro

VPC ID
 vpc-0a7f5e6a893982551 (Network_vpc)

Subnet ID
 subnet-0a43a8db4778da223 (public1)

Instance ARN
 arn:aws:ec2:us-east-1:531472034848:instance/i-0832733fd8f8b465b

Private IPv4 addresses
 10.0.1.109

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
 Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

8- create ec2(application) private subnet with security group from 8

```
1
2 resource "aws_instance" "app" {
3     ami                    = "ami-0c02fb55956c7d316"
4     instance_type         = "t3.micro"
5     subnet_id             = aws_subnet.private1.id
6     vpc_security_group_ids = [aws_security_group.allow-ssh-port_3000.id]
7     associate_public_ip_address = true
8     key_name              = "terr"
9
10    tags = {
11        Name = "app"
12    }
13 }
```

Instance summary for i-Od5c4ba5720b548a5 (app) Info

Updated less than a minute ago

Instance ID

 i-Od5c4ba5720b548a5

IPv6 address

-

Hostname type

IP name: ip-10-0-3-190.ec2.internal

Answer private resource DNS name

-

Auto-assigned IP address

 18.215.63.8 [Public IP]

IAM role

-

IMDSv2

Optional
 EC2 recommends setting IMDSv2 to required | [Learn more](#) 

Operator

-

Public IPv4 address

 18.215.63.8 | [open address](#) 

Instance state

 Running

Private IP DNS name (IPv4 only)

 ip-10-0-3-190.ec2.internal

Instance type

t3.micro

VPC ID

 vpc-Oa7f5e6a895982551 (Network_vpc) 

Subnet ID

 subnet-Oe22020712bcb3b68 (private1) 

Instance ARN

 arn:aws:ec2:us-east-1:531472034848:instance/i-Od5c4ba5720b548a5

Private IPv4 addresses

 10.0.3.190

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#) 

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Security groups

 sg-056becc7fedb684fe (allow_ssh and port 3000 only)

Owner ID

 531472034848

Launch time

Fri May 01 2026 16:10:52 GMT+0300 (Eastern European Summer Time)